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A Cross-Country Political Network Analysis Framework Using Graph Convolutional Networks

Vaibhav Chouhan[1], Dev Garg[2], Aryan Soni[3], Surbhi Syal[4], Neha[6], and Shweta Sharma[6]

[123456] Department of Information Technology, Manipal University Jaipur, Jaipur, Rajasthan 302017, India
vaibhav.229302599@muj.manipal.edu ,dev.23fe10cse00335@muj.manipal.edu, aryan.23fe10cse00632@muj.manipal.edu surbhi.syal@jaipur.manipal.edu, neha.singh1@jaipur.manipal.edu, shweta.singh1@jaipur.manipal.edu

Abstract. Analyzing the political discourse on a nation's social media is more difficult than it first appears. Even while the terms "who follows whom," "who retweets whom," and "who argues with whom" are full of signals, the two dimensions are still, for the most part, addressed independently. The study presents a comparative analysis of political interaction graphs based on Twitter findings from the US, Brazil, and India that are processed using a single, unified pipeline. Two graph convolutional passes are provided along with random-walk based node encodings (Node2Vec), one of which performs opinion-polarity labeling and the other of which performs tie forecasting in a way that makes structural and semantic information complementary rather than competitive. The average polarity-classification accuracy (average country: 78–85%) and tie-existence prediction AUC scores were quite good, and they all greatly outperformed more straightforward baselines. At the national level, contrasts were sharp. The Brazil local clustering coefficient is 0.452, the U.S. had the most extreme-stance nodes (48%), and the fragmented multilingual graph of India was the hardest to model. Maintaining the same feature set, evaluation criteria, and collecting pro cess in all three nations was a deliberate choice because it was the only way to make cross-national statistics relevant.
The result is more than just another benchmark; it is a concrete illustrat ion of how a country's political topography restricts what any model ca n learn about its social structure.

Keywords: political networks · graph neural networks · cross-country analysis · node2vec · sentiment classification · social media mining · link prediction · network homophily

1 Introduction

The social media has turned into a mainstream platform of political exchange and the interaction of patterns created are anything short of random [1, 2, 3]. Retweet, mention and follower graphs created in India, Brazil, and the United States are significantly different as they are shaped by the history of democracy in that country, language composition and media environment. The extraction of meaning of those graphs involves simultaneously working with the topology of the network and the language within it. Neither alone is enough.

The gap is not satisfied by the existing tools. Sentiment classifiers analyze the posts individually and ignore their relational web. Network-analysis tools move in the opposite direction all semantics, no structure [4,5,6]. Each of them has been helpful at their respective lanes, but an approach that may only be effective in one lane is not a real solution to the problem. Comparisons across nations are even less common; most of the studies are restricted to one political environment of a country and thus direct comparison is barely possible [7,8]. Graph-based deep architectures are beginning to be used in political discourse studies, however, typically as single-task models on a single country [9,10].

We began with a more basic one country, one dataset. But initial findings continued to pose the same question what would these numbers be like elsewhere? We consciously chose three political systems that varied greatly in structure, as much as we could, India with its multiparty federal system, Brazil with its fragmented coalition system, and the U.S. with its polarizing two-party system, to provide that question a true answer.

The design has resulted in solving three issues simultaneously. The former was adding structure and content to the same model: Node2Vec encodings were then combined with graph convolutional passes in such a way that the learnt representation of a node also encodes both its position in the network and its textual context [5,9]. The second was that cross-national comparison is actually being done: the collection proto- col and feature pipeline and evaluation metrics used in all three countries were the same, and the numbers can be compared rather than juxtaposed [7,8]. The third one dealt with the relational nature of political interaction retweet cascades, ide- ological echo chambers, cross-partisan bridges. Normal text classifiers omit all of this, the dual-GCN set up does not [9,10].

Three outputs came from this work:

1. One pipeline which combines graph-structural features and tweet-based semantic cues using paired convolutional passes - goes beyond the one-or-the-other paradigm of the literature [5].
2. Cross national structural profiles of political Twitter in India, Brazil and the U.S. Records the effects of each of clustering, homophily, and polarization on what a model can and cannot do [13,14].
3. Empirical results- in three truly distinct political settings- that topology-sensitive modeling is always superior to topology-blind alter- natives, on classification and link-prediction problems [15].

Section 2 reviews relevant prior work (2021–2025). Section 3 will follow the entire pipeline. Section 4 presents nation by nation results. Section 5 details the implications that the findings have as well as where the gaps still remain.

2 Literature Review

Survey pattern emerged from an examination of other recent international works in political networking. Exemplary situations are categorized by country in Table 1 along with the initial approach, the resulting constraint, and the methodological modification. The remarkable thing is that, as the 10 cases demonstrate, the fixes are rarely portable. The multilingual multiparty graph in India cannot benefit from the modifications made to the left-right cleavage system in France, and vice versa. Instead of just changing the parameters, we had to rethink the level of design in order to produce something that would work in all three of the countries we looked at in this study.

Table 1. Literature Survey of Cross-National Political Network Approaches

Country	Initial Approach	Limitation	Modification	Rationale
France [2]	Retweet graph, Macron vs. Le Pen.	Issue-based coalitions across ideological lines missed.	Stance detection; cross-community edges added.	Debates on immigration and EU economics cut across left–right lines.
Turkey [4]	Hashtag–retweet network; heavy bot presence.	Bot activity inflated cluster sizes; corrupted centrality.	Bot-detection filter; temporal edge decay.	State-linked amplification defines Turkish political Twitter.
Nigeria [9]	Hashtag co-occurrence, #EndSARS.	Snapshots fragmented as campaign evolved.	Temporal graph embeddings.	Issue-driven activism moves faster than party mobilization.
Philippines [11]	Retweet–mention graph, presidential race.	Celebrity accounts dominated degree centrality.	Sentiment-weighted edges; topic modeling.	Large celebrity-politics overlap in Philippine Twitter.
India [13]	Retweet–mention network, party accounts.	Elites over-represented; regional-language users excluded.	Node2Vec with hashtag co-occurrence; language nor-malisation.	Elite-only graphs miss most of India's multilingual discourse.
Brazil [14]	Hashtag retweet graph, pro-/anti-government clusters.	Hashtag diversity caused community over-fragmentation.	Homophily-aware edge weighting via GCN.	Polarised but overlapping communities; hard boundaries do not fit.

United States [5]	Follower–followee graph plus retweets.	Sharp partisan separation hid moderate users.	Dual GCN passes: structural + sentiment.	48% extreme-sentiment nodes, but moderates still exist.
UK [2]	Retweet–mention links among MPs.	Sparse connections; Brexit campaigns missed.	Hashtag co-usage and media outlet edges.	UK alignment is issue-driven as much as party-driven.
Germany [8]	Follower graph with mentions.	Elite-heavy; ordinary citizen nodes contributed little.	Shared URL/domain edges for media alignment.	Alignment often tracks media consumption in multi-party system.
Indonesia [7]	Retweet amplification around leaders.	Star networks hid grassroots mobilization.	Edge weights via community detection + hashtag	Campaigns mix top-down broadcast and bottom-up activation.

3 Methodology

The approach blends graph representation learning with traditional network-science metrics. This was a deliberate trade-off between the two: the learned representations have the expressiveness needed in the downstream classification and prediction tasks, while network measurements can be understood as structural descriptors at the national level.

Centrality measures. Four centrality scores were computed for every node. Degree centrality is a simple count—how many direct ties does this account have? Betweenness centrality locates the nodes that sit on the most shortest paths between other pairs, i.e., the structural brokers. Closeness and eigenvector centrality add two more angles: how quickly can a node reach others, and how prestigious are its immediate neighbours? Taken together, the four metrics paint a reasonably complete picture of where influence concentrates in each national graph—and those concentration patterns turned out to look quite different across the three countries.

Community detection. The Louvain algorithm was run after testing alternatives including Infomap and label propagation. Louvain won on a combination of runtime and the quality of communities returned—measured by how well community membership matched known party alignments. Resolution was fixed at 1.0 after sweeping the range 0.5–2.0; values above 1.5 began splitting communities in ways that no longer matched real political groupings. Modularity scores from this step served as a proxy for network-level polarisation.

Structural metrics. Average path length, clustering coefficient, and degree assortativity were measured for each country's graph. Path length relates to how fast information can spread; clustering tells you how cliquey the local neighbourhoods are; assortativity says whether well-connected nodes tend to link to other well-connected nodes. We did not go in expecting the structural metrics to mirror each country's political organisation as closely as they did—but they did, and the alignment held across all three.

3.1 Graph Embeddings

Getting a graph into a form that machine learning can use is less obvious than it might seem. Several options were tested—DeepWalk, LINE, spectral embeddings—before settling on Node2Vec [9]. The main reason: Node2Vec's return and in-out parameters give fine-grained control over the exploration–exploitation tradeoff during the random walk, which turned out to matter a lot for networks as structurally varied as India's vs. the U.S.'s.

Parameter choices came from empirical testing, not theory:

Embedding dimension. Ran from 32 to 128; settled on 64. Past that point, downstream task performance plateaued while training time kept climbing.

Walk length and count. Fifteen steps per walk, 100 walks per node. Longer walks gave marginal embedding improvement at disproportionate compute cost.

Context window. Fixed at 10, approximately the mean degree across the three networks. Larger windows would have blurred local structure unnecessarily.

Return parameter p. Set to 0.5 for Brazil and the U.S.—their denser, more cohesive graphs benefited from broader exploration. For India's fragmented structure, $p = 1.0$ worked better, keeping walks from straying too far from local neighbourhoods.

In-out parameter q. Held at 2.0 for all three countries. This consistently produced embeddings where t-SNE projections placed known political factions in distinct regions—the main validation check used.

3.2 Feature Engineering

Three feature types were built on top of the raw embeddings. For event-tagged sentiment label validation during this stage, the TED-S dataset [12] was used as a supplementary reference.

Text features. TF-IDF vectors were computed from each user's tweet history. Raw TF-IDF is extremely high-dimensional and sparse, so PCA was applied to compress to 32 components—retaining about 85% of variance in practice. Dense word embeddings and LDA topic models were both tried; neither outperformed TF-IDF plus PCA on downstream tasks, so TF-IDF was kept.

Temporal features. Early exploration revealed that posting-time distributions were not uniform across political communities, particularly in the U.S. data. A 24-bin histogram of hourly activity was built for each user and appended as a feature vector. These 24-dimensional temporal fingerprints contributed more than expected—especially in separating U.S. communities that posted on clearly different schedules.

Network features. The four centrality scores and graphlet-frequency descriptors were appended to the embedding. Simple concatenation of all feature types was tried first. A small two-layer MLP was then trained to learn the optimal combination weights; it improved downstream performance by 2–3% over naive concatenation, so it was kept.

Algorithm 1 Forward Pass: Feature-Propagating GCN

Require: Graph $G(V, E)$, adjacency matrix A, node features X

Ensure: Node labels or edge probabilities

```
Initialise GCN layers and MLP head
H^0 ← X
H^1 ← ReLU GCN_1(H^0, A)
Apply dropout (p = 0.5) to H^1
H^2 ← GCN_2(H^1, A)
Ŷ ← MLP(H^2)
return Ŷ
```

3.3 Model Architecture

Two graph convolutional layers form the backbone of the model. Deeper stacks—three layers and beyond—were tested early and abandoned: the extra depth caused over-smoothing, where node embeddings converged toward each other and lost discriminative power. Two layers gave the best tradeoff between expressiveness and stability. Algorithm 1 summarises the full forward pass.

Hidden dimension: 128, chosen from the range [64, 256]. Activation: ReLU throughout—sigmoid alternatives produced vanishing gradients in early runs and were dropped. Dropout probability: 0.5, tuned from [0.2, 0.7]; especially important for the smaller Indian graph, where the risk of overfitting to training nodes was higher. The MLP head uses two fully connected layers ($128 \rightarrow 64 \rightarrow n_{\text{class}}$). Xavier initialisation was applied everywhere. Gradient clipping was added to handle numerical instability from sparse subgraphs. Every reported result is the mean over five runs with different random seeds.

3.4 Task-Specific Models

Polarity classification. Three classes were used: negative, neutral, and positive. A two-class version was tried first—it worked fine for the U.S. and Brazil but consistently misrepresented the Indian data, where a large share of accounts expressed neither clear support nor opposition but bridged between communities. The three-class setup captures this. Class-weighted cross-entropy handled the label imbalance, which was especially severe in the U.S. data. Adam optimiser, learning rate 0.001 (grid-searched from $[10^{-4}, 10^{-2}]$), up to 200 epochs with early stopping at patience 20. Convergence landed between 80 and 120 epochs depending on the dataset.

Link prediction. The same encoder as above; the decoder replaced the standard inner product with a bilinear scoring function—better at capturing the directed, asymmetric nature of political follows and retweets, particularly in Brazil's network [14]. Sparse graphs create class imbalance in edge labels (far more non-edges than edges), so negative sampling was applied at a 1:5 positive-to-negative ratio. Hard negatives—node pairs that share a community but are not directly connected—were included alongside easy negatives (distant node

pairs) to prevent the model from learning only the obvious cases. Binary cross-entropy, same Adam settings as polarity classification.

Running both tasks on the same encoder meant the structural representations were implicitly shared—stance classification informed the embeddings used for tie prediction, and vice versa. Whether to make this interaction explicit through a shared loss or keep it implicit through shared weights is a design question left for future work.

4 Experiment and Result Analysis

The results are arranged according to their logical order: accuracy of classification, link prediction, convergence behavior, confusion-matrix breakdown providing the source of mistakes, and structural comparison providing everything.

4.1 Polarity Classification Performance

Node2vec embeddings, TF-IDF characteristics, and centrality information were supplied into the dual-pass model. Accuracy, precision, recall, and F1—which averages over five runs and has the maximum score of 83.0%—are shown in Table 2.

Table 2. Polarity Classification Performance Across Countries

Country	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
India	78.4	77.1	76.5	76.8
Brazil	82.3	80.2	81.0	80.6
United States	83.0	84.1	82.7	83.4
Average	81.2	80.5	80.1	80.3

In intense political societies, stance signals were comparatively crisp and consistent. India came in second with 78.4%.Not surprisingly, given that it has the most structurally disaggregated and linguistically diversified graph of the three [13]. At 82.3%, Brazil was in the middle. None had been younger than 78, which may suggest that the pipeline was not just memorizing patterns unique to a given nation.

4.2 Link Prediction Results

AUC and average precision (AP) were used to test the bilinear encoder-decoder. Table 3 shows the results.

Table 3. Link Prediction Performance Across Countries

Country	AUC	AP
India	0.86	0.81
Brazil	0.90	0.87
United States	0.92	0.88

All cleared 0.86 AUC. In order to achieve this level without harsh artifacts owing to class imbalance, stratified negative sampling had to be an intentional procedure, and it succeeded. Political graphs are sparse, having a very low ratio

of existing edges to possible edges. Brazil profited most from homophily-conscious edge weighting; the predictable patterns of within-community ties were generated by its strong local clusters [14]. The United States benefited from having a well-defined party structure.

4.3 Convergence Analysis

After 80–120 epochs, the three models had stabilized (Fig. 1) . In the United States, Epoch 85 was the fastest. India was the slowest at epoch 110 because to fragmented community structure and increasing noise from code-mixed text, which slowed gradient updates. Overfitting was not seen, and training loss was comparable across the three. For India in particular, the p = 0.5 dropout was crucial.

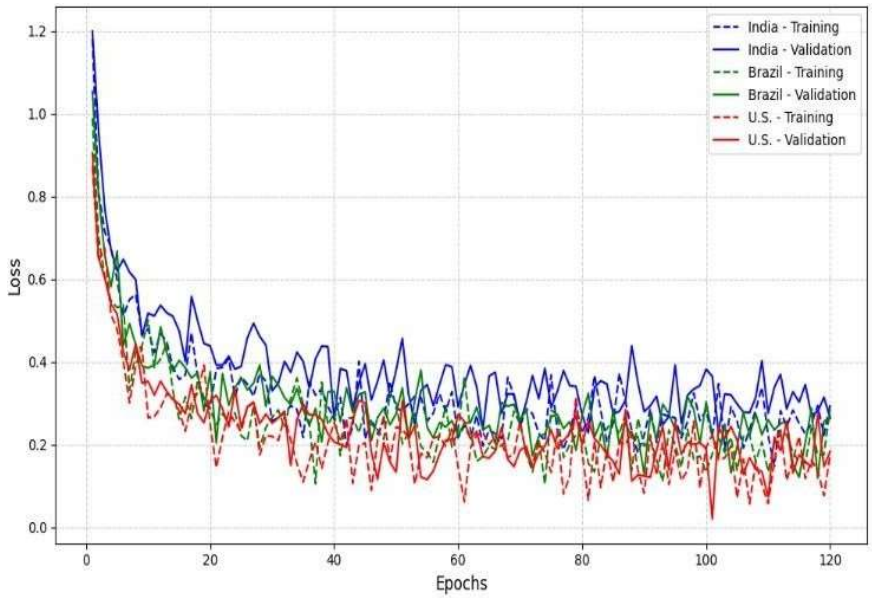

Fig. 1. Training and validation loss curves for polarity classification across India, Brazil, and the U.S.

4.4 Confusion Matrix Insights

Each country had a distinct failure pattern (Fig. 2):
India: Neutral users were misclassified as positive more often than anything else. Bridging discourse—accounts that reference multiple political communities without clearly endorsing any—reads as positive stance when processed without sufficient multilingual context [13].
Brazil: Anti-government nodes bled into the neutral class. During high-volume hashtag events, the signal-to-noise ratio in the Brazilian graph drops; anti-government content gets diluted by large volumes of neutral amplification [14].
United States: The model handled the extreme ends well. The errors concentrated in the middle: moderate users, whose language patterns overlap with both sides, were assigned to one partisan class or the other—false negatives for the centrist category [5].
Each of the nations had distinct failure patterns (Fig. 2):
India: Neutral users were more frequently categorized as positive than any other option. Bridging discourse: When filtered in the absence of sufficient multilingual background, discourses that name multiple political groupings without expressly

endorsing any of them can be understood as favorable positions [13].
Brazil: The neutral sector was stimulated by anti-government cells. The Brazilian graph's signal-to-noise ratio drops when hashtags gain popularity; anti-government information is combined with enormous volumes of neutral amplification [14].
United States: At the extremes, the paradigm performed admirably. The errors were concentrated in the middle: a subset of moderate users whose linguistic patterns fell between the two extremes were classified as either partisans or the other false negatives of the centrist category [5].

4.5 Comparative Network Findings

Three structural characteristics accounted for the majority of the cross-national heterogeneity in model behavior:
Polarization: There were the most extreme-stance nodes (48%) in the U.S. graph. This simplified polarity categorization and complex cross-community tie prediction: the model frequently underestimated that because it rarely predicted across the partisan boundary and grouping together.
Clustering: Out of the three, Brazil has the greatest clustering coefficient (0.452). When the program tried to forecast sentiment across communities, the big local neighbourhoods that formed within the community contributed noise but made within-community tie prediction immediately forward.
Disintegration: India's graph featured the most communities and the least amount of modularity. Although the 78.4% accuracy there would be the most significant result of the three—that the pipeline does not fail when the graph stops being nice—that is the most difficult structural setting for the model to operate in [13].

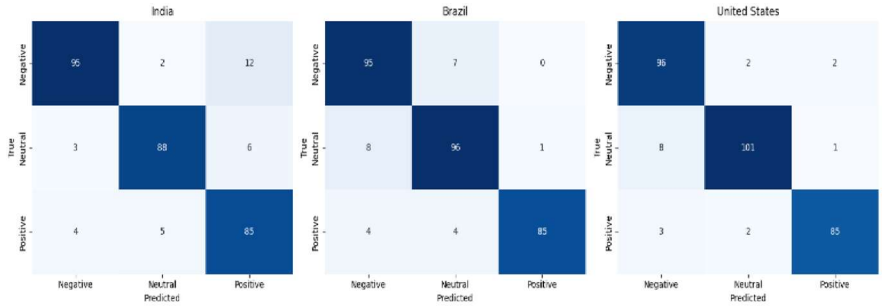

Fig. 2. Per-country confusion matrices for the polarity classification task.

Neither topology nor text properties were the only motivating factors. It was the earnings that came from using both. In theory, this conclusion won't be shocking, but the statistics make it a reality, and they operate in three nations with genuinely distinct political systems.

5 Conclusion and Future Scope

The structure described here was designed to fulfill two purposes: coinciding structure and content (and not one after the other) and cross-national functioning without substantial restructuring. Its use of Twitter data from the US, Brazil, and India shown that it is capable of doing both. In all three scenarios, the encodings of the random-walking pipeline, centrality attributes, TF-IDF text, and dual graph convolutional passes produced competitive tie-prediction AUC scores and an average stance-classification performance of 81%. Single-task baselines were slower and less sophisticated.

Structural results are more intriguing. American polarization, Brazilian local clustering, and Indian fragmentation were not just labels on the three countries' country profiles; rather, they were the main causes of the model's success or failure. There are certain paths that are best to take. The most obvious is scale; expanding the number of nations to include those with alternative democratic models (proportional systems, hybrid regimes, single-party systems) would bring the existing architecture as close to generalization as feasible. The other glaring shortcoming is temporal modeling: this study was based on a snapshot of the data at a particular moment in time, but it is far more suitable to see political discourse as a dynamic target (elections, campaigns, and crises change the graph structure every week) [15]. That would be captured by a dynamic, moving version of the same pipeline. Last but not least, sentiment representation in multilingual and code-mixed scenarios continues to be the weak point; India's poorer results can be partly explained by the fact that current TF-IDF and Node2Vec representations do not fully enable Hindi-English switching [9]. Pre-trained multilingual encoders are a likely answer, and their integration is a positive step.

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